Library Management System

Java OOP Console Application — Project Report

About the Project

A menu-driven Library Management System built in Java using core Object-Oriented Programming concepts. The program runs in a loop and lets the user add books, search them, issue and return them — just like a real library counter. All classes are in a single .java file, making it easy to compile and run.

Program Structure

Class	Field / Method	Description
Book	String title	Name of the book
	String author	Author's name
	boolean isAvailable	true = Available, false = Issued
	Book(title, author)	Constructor — initializes book, availability = true
	showDetails()	Prints title, author, status
Library	Book[] books	Array to store all Book objects
	int count	Tracks total books added
	Library(int size)	Constructor — creates array with given capacity
	addBook(Book b)	Adds a new book to the array
	searchBook(String title)	Case-insensitive search, returns Book or null
	issueBook(String title)	Issues the book if available
	returnBook(String title)	Returns the book, updates status
	showAllBooks()	Displays all books with status
LibraryManagement	main(String[] args)	Scanner input + menu loop (do-while)

OOP Pillars Used

Pillar	Used?	Where
Encapsulation	Yes	Data and methods bundled inside Book and Library class
Abstraction	Yes	User calls issueBook(), inner logic is hidden
Inheritance	No	Not needed for this scope

Pillar	Used?	Where
Polymorphism	Partial	showDetails() works differently on each Book object

Features

• Start with empty library — no hardcoded books

• Add books manually at runtime via Scanner input

• Search book by title (case-insensitive match)

• Issue a book — marks it as Issued

• Return a book — marks it back as Available

• View all books with their current status

• Handles all edge cases (book not found, already issued, not issued)

• Menu runs in a do-while loop until user selects Exit

How to Run

Step 1 — Compile the file:

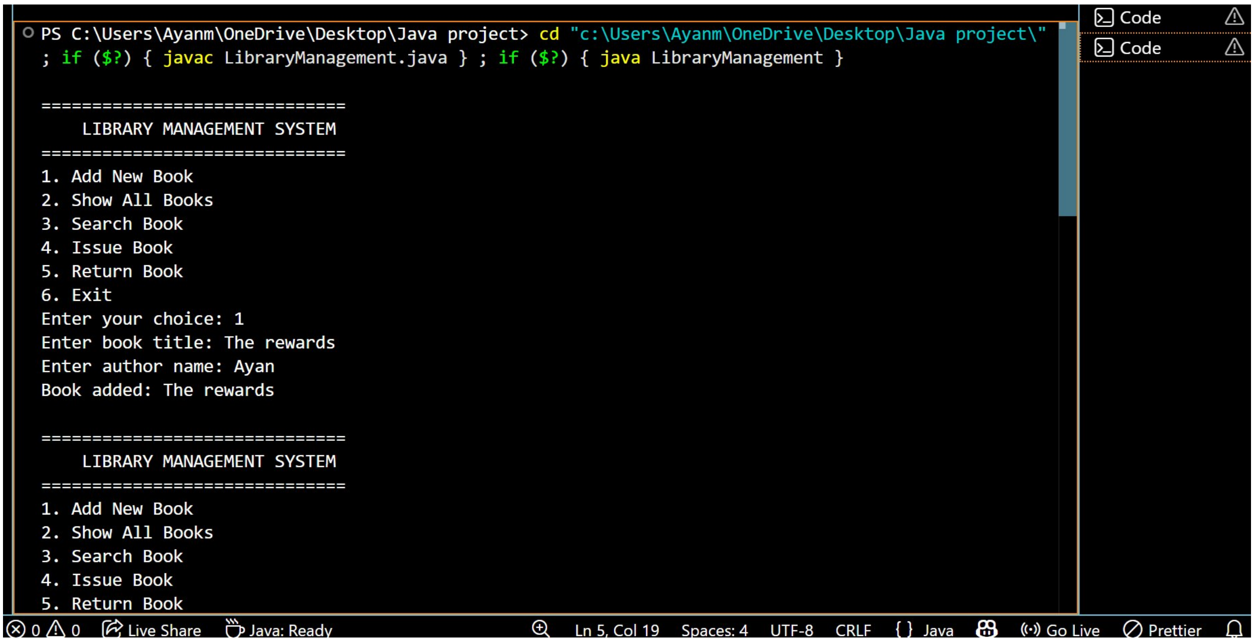
```
javac LibraryManagement.java
```

Step 2 — Run the program:

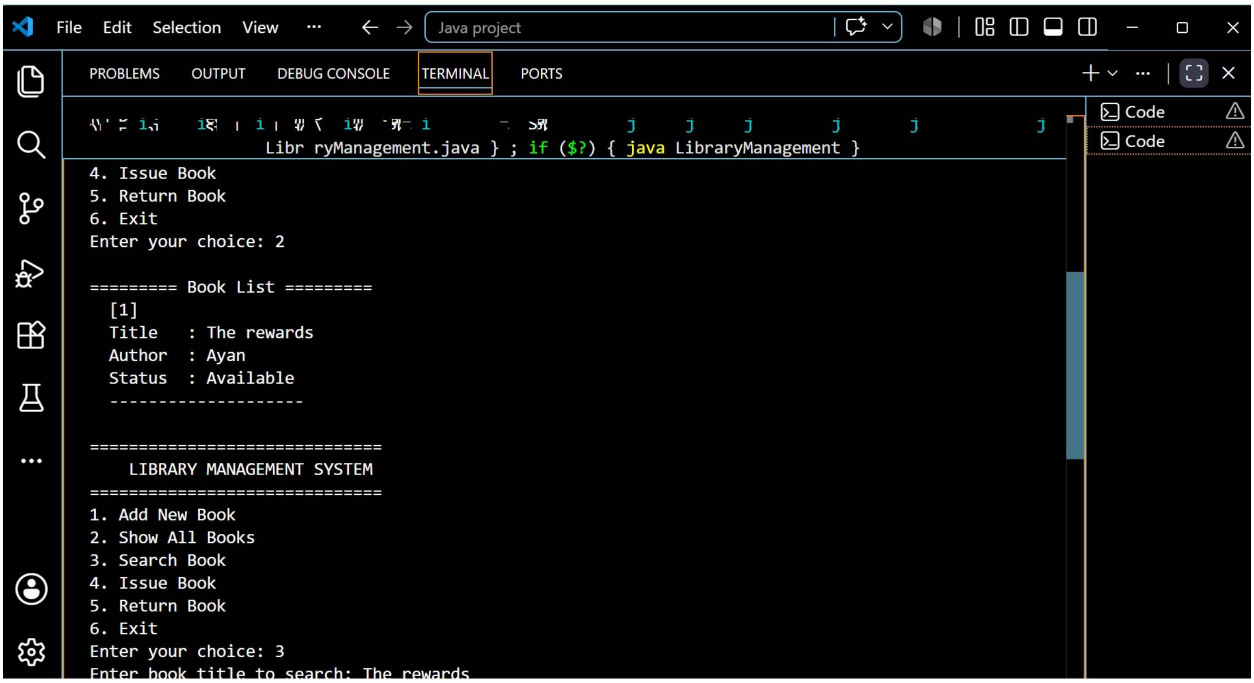
```
java LibraryManagement
```

Requires Java 8 or above. No external libraries needed.

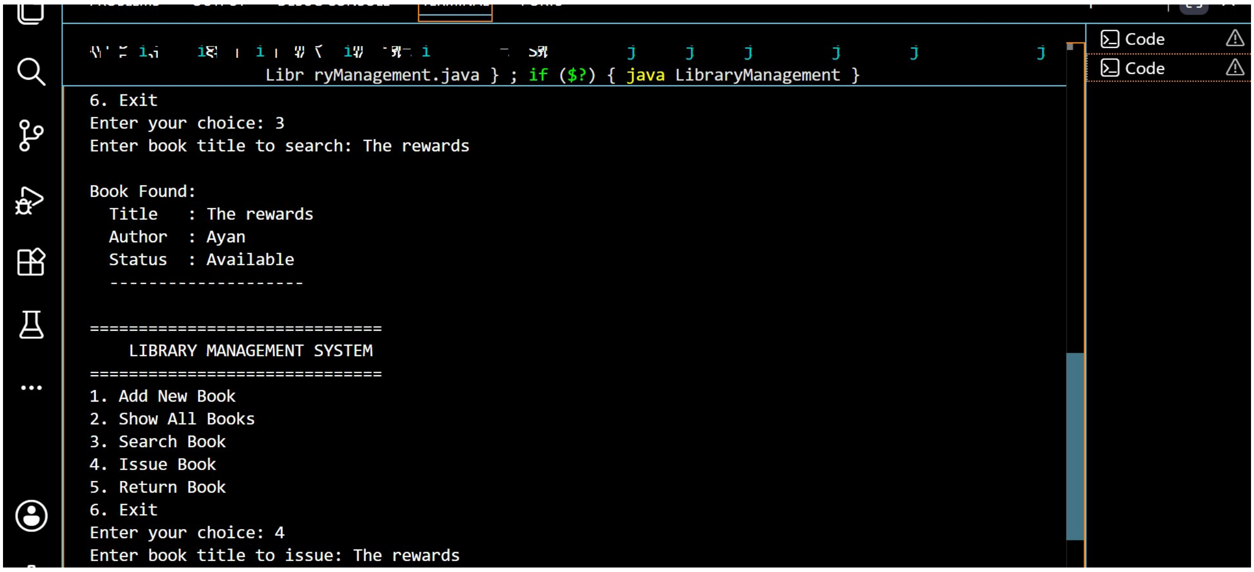
Program Output — Screenshots



Screenshot 1: Program starts — menu displayed, user adds book 'The rewards' by Ayan



Screenshot 2: Show All Books (Option 2) and Search Book (Option 3) in action



Screenshot 3: Search result displayed, then Issue Book (Option 4) executed

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